

Stability analysis of systems with uncertain time-varying delays[☆]

Chung-Yao Kao^{a,*,1}, Anders Rantzer^{b,2}

^aDepartment of Electrical and Electronic Engineering, University of Melbourne, Parkville, Vic., Australia

^bDepartment of Automatic Control, LTH, Lund University, Lund, Sweden

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Abstract

Stability of systems in the presence of bounded uncertain time-varying delays in the feedback loop is studied. The delay parameter is assumed to be an unknown time-varying function for which the upper bounds on the magnitude and the variation are given. The stability problem is treated in the integral quadratic constraint (IQC) framework. Criteria for verifying robust stability are formulated as feasibility problems over a set of frequency-dependent linear matrix inequalities. The criteria can be equivalently formulated as semi-definite programs (SDP) using Kalman–Yakubovich–Popov lemma. As such, checking robust stability can be performed in a computationally efficient fashion.

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1. Introduction

It is well known that the manifestation of time delays in a system can lead to performance degradation and even destabilization of the system. For systems in which the signal transmission time delays among sensors, compensators, and actuators are small compared to the time constant of the overall system, the effect of the time delays is often not significant enough to cause serious problems. This is not the case, however, within the context of large-scale distributed and networked systems, where the effect of time delays can be very significant. For such systems, time-delay robustness must be explicitly addressed to ensure that system-level performance is achieved in a robust manner (Zhang, Branicky, & Phillips, 2001).

Time-delay robustness is often studied for situations in which the delays are uncertain but remain constant throughout time. While much research has been done and stability criteria have been derived for systems with uncertain constant time delays,

the recent emphasis has been put on the scenario where the time delays are time-varying. The significance of such problems is tied to the recent ample interest in designing control algorithms for large-scale networked systems. For applications within the scope of large-scale distributed and networked systems, such as the regulation of internet traffic and control over networked communication channels (Low, Paganini, & Doyle, 2002; Misra, Gong, & Towsley, 2002), the network-induced time delays are by nature time-varying, sometimes randomized, and often the variation is significant enough not to be ignored. Other engineering applications where time-varying delays appear include real-time implementation of control systems and control of fuel injection systems (Cervin, Henriksson, Lincoln, & Årzén, 2003; Cho & Hedrick, 1989).

Consider the following linear time-delay system:

$$\dot{x}(t) = Ax(t) + A_d x(t - \tau(t)), \quad (1)$$

where $\tau(t)$ is an unknown time-varying parameter satisfying

$$0 \leq \tau(t) \leq \tau_0, \quad |\dot{\tau}(t)| \leq d, \quad \forall t \geq 0, \quad (2)$$

A and $A_d \in \mathbf{R}^{n \times n}$ are constant matrices. The initial condition $x(\theta)$ is a continuous function defined on $[-\tau_0, 0]$. In this paper, we are interested in delay-dependent conditions for robust stability of the time-delay system (1). More specifically, given a pair of scalars (τ_0, d) , our objective is to derive conditions under

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* Corresponding author. Tel.: +61 3 83444685; fax: +61 3 83446678.

E-mail address: cykao@ee.unimelb.edu.au (C.-Y. Kao).

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which the delay system (1) is stable for all $\tau(t)$ that satisfy condition (2).

If the delay parameter τ is unknown but constant, then the energy of $x(t - \tau)$ is the same as the energy of $x(t)$. Hence, a simple but conservative delay-independent stability criterion for the system, $\sup_{\omega} \| (j\omega I_n - A)^{-1} A_d \| < 1$, immediately follows the small gain theorem. The exact condition for delay-independent stability was derived in Chen and Latchman (1995) using structured singular value. Ever since, the focus has been put on the delay-dependent stability. While some of them were derived using frequency-domain analysis (by μ analysis or IQC analysis) (Fu, Li, & Niculescu, 1998; Huang & Zhou, 2000; Jun & Safonov, 2001, 2002; Megretski & Rantzer, 1997; Morari & Zafriou, 1989; Wang, Lundström, & Skogestad, 1994; Zhang, Knope, & Tsiotras, 2001), many others use time-domain analysis; namely, the Lyapunov second method. See, for example, Li and de Souza (1997b), Park (1999), Kolmanovskii and Richard (1999), Han and Gu (2001), Xu and Lam (2005), Zhang, Knope, and Tsiotras (2005). Interested readers are referred to Kolmanovskii, Niculescu, and Gu (1999) and the research monograph by Niculescu (2001) for references and the recent development on analysis of time-delay systems.

When the delay parameter is time-varying, stability analysis is more involved. One of the difficulties, for instance, is that the delay operator is no longer energy-preserving. In fact, if there is no restriction on the variation, i.e., no bound on $\dot{\tau}(t)$, then the delay operator is not even a bounded operator on the L_2 space no matter how small the length of the delay is. To see this, let $v(t)$ and $\tau(t)$ be

$$v(t) = \begin{cases} 1, & t = [0, \varepsilon], \\ 0 & \text{otherwise,} \end{cases} \quad \tau(t) = \begin{cases} t, & t \in [0, \tau_0], \\ \tau_0 & \text{otherwise.} \end{cases}$$

Then, $v(t - \tau(t))$ is equal to 1 for $t \in [0, \tau_0 + \varepsilon]$ and 0 otherwise. The energy of $v(t - \tau(t))$ is equal to $\tau_0 + \varepsilon$ while the energy of $v(t)$ is equal to ε . Hence, the gain of the delay operator becomes unbounded as $\varepsilon \rightarrow 0$. Therefore, systems with time-varying delays are intuitively less stable than those with constant time delays, and it is not obvious that stability criteria for systems with constant time delays can be easily generalized for time-varying delay systems. We have observed extensive research activities on the topic of robust stability or stabilization of systems with time-varying delays over the past few years, among which the conditions for stability are mostly derived by the time-domain (Lyapunov) approach, based on various Lyapunov–Razumikhin functions or Lyapunov–Krasovskii functionals. While a comprehensive literature review is beyond the scope of this paper, interested readers are referred to (Cao, Sun, & Cheng, 1998; Fridman & Shaked, 2002, 2003; Gu & Han, 2000; Kharitonov & Niculescu, 2003; Kim, 2001; Li & de Souza, 1997a; Mehdi, Boukas, & Liu, 2002; Suplin, Fridman, & Shaked, 2006; Wu, He, She, & Liu, 2004) for a (partial) development of the field.

In this paper, we consider the stability problem where the delay parameter is bounded but time-varying. We adapt the frequency-domain approach to attack the problem. Specifically, the stability problem is treated in the integral quadratic constraint (IQC) framework (Megretski & Rantzer, 1997). The

main advantage of IQC analysis is that the result obtained can be easily generalized to systems with delays and uncertainties (e.g., unknown parameters or unmodelled dynamics) or nonlinearities (e.g., saturation or dead zone). To our best knowledge, the frequency-domain approach for varying time-delay robustness was not seriously attempted before. Ideas in this direction can be found in Gu, Kharitonov, and Chen (2003), our preliminary results published in Kao and Lincoln (2004) and Kao and Rantzer (2005), and a recent paper by Fridman and Shaked (2005).

Notation Symbol I_n is used to denote n -dimensional identity matrix. The subscript n is dropped when the dimension is evident from the text. Given a matrix M , the transposition and the conjugate transposition are denoted by M' and M^* , respectively. The notation $M > 0$ (“ $\geq$ ”, “ $<$ ”, and “ $\leq$ ”, respectively) is used to denote positive definiteness (positive semi-definiteness, negative definiteness, and negative semi-definiteness, respectively). Symbol L_2^m denotes the space of $\mathbf{R}^m$ -valued, square summable functions defined on time interval $(-\infty, \infty)$, and L_{2e}^m denotes the extension of the space L_2^m , which consists of functions whose time truncation lies in L_2^m . Notation $\mathbf{RL}_{\infty}^{l \times m}$ is used to denote the space of proper rational transfer matrices (of dimension $l \times m$) with no poles on the imaginary axis, while $\mathbf{RH}_{\infty}^{l \times m}$ denotes the subspace of $\mathbf{RL}_{\infty}^{l \times m}$ consisting of functions which have no poles in the closed right half plane. Every $H \in \mathbf{RL}_{\infty}^{l \times m}$ defines a convolution operator on L_2 : let $h(t)$ be the inverse Laplace transform of H . Then, for any $u \in L_2$,

$$(Hu)(t) := \int_{-\infty}^{\infty} h(t - \theta)u(\theta) d\theta.$$

Given a signal f in the L_2 space, we use $\|f\|_{L_2}$ to denote the L_2 norm of f . Given a bounded operator G on the L_2 space, we use $\|G\|_{L_2}$ to denote the L_2 -induced norm of G .

Given a signal $f(t)$, the Fourier transform of the signal is denoted by $\hat{f}(j\omega)$. The closure of the set $\{t | f(t) \neq 0\}$ is called the support of f and is denoted by $\text{Supp}(f)$.

Let $\mathcal{D}_{\tau}$ denote the time-delay operator $\mathcal{D}_{\tau}(v) := v(t - \tau(t))$, and $\mathcal{S}_{\tau}$ be $(I - \mathcal{D}_{\tau})$; i.e., $\mathcal{S}_{\tau}(v) := v(t) - v(t - \tau(t))$. To simplify the notation, in the rest of the paper we will suppress the time dependency on $\tau(t)$ and simply write τ and $\dot{\tau}$.

2. Stability analysis via IQC

In this section, we briefly introduce the IQC analysis, which is needed for the main result of this paper. The system under consideration is

$$v = Gw + e, \quad w = \Delta(v), \quad (3)$$

where $G(s) \in \mathbf{RH}_{\infty}^{l \times m}$, $e \in L_{2e}^l$, and Δ is a bounded and causal operator on L_{2e}^l . *Well-posedness* and *stability* of such a system are defined as follows.

Definition 1 (Megretski & Rantzer, 1997). The feedback interconnection of G and Δ as defined in equation (3) is said to be *well-posed* if the map $(v, w) \mapsto e$ has a causal inverse on L_{2e} . That is, for any $e \in L_{2e}$, there exists a solution $(v, w) \in L_{2e}$

which depends causally on e . If, in addition, there exists a positive constant C such that

$$\int_{-\infty}^T \|v\|^2 + \|w\|^2 dt \leq C \int_{-\infty}^T \|e\|^2 dt, \quad \forall T \geq 0$$

then we say the system is stable.

Let Π be a bounded self-adjoint operator on the L_2 space. Then Π defines a quadratic form on L_2

$$\begin{aligned} \sigma_{\Pi}(v, w) &:= \left\langle \begin{bmatrix} v \\ w \end{bmatrix}, \Pi \begin{bmatrix} v \\ w \end{bmatrix} \right\rangle \\ &= \int_{-\infty}^{\infty} \begin{bmatrix} v(t) \\ w(t) \end{bmatrix}' \left(\Pi \begin{bmatrix} v(t) \\ w(t) \end{bmatrix} \right) dt \\ &= \int_{-\infty}^{\infty} \begin{bmatrix} \hat{v}(j\omega) \\ \hat{w}(j\omega) \end{bmatrix}^* \Pi(j\omega) \begin{bmatrix} \hat{v}(j\omega) \\ \hat{w}(j\omega) \end{bmatrix} d\omega. \end{aligned}$$

The operator Π is referred to as the multiplier of the quadratic form σ_{Π} . To facilitate the development, the multiplier Π is often block decomposed into the form

$$\begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{12}^* & \Pi_{22} \end{bmatrix},$$

where the dimensions of Π_{ij} confirm to those of v and w .

Given an operator $\mathcal{H}$ and a quadratic form $\sigma_{\Pi}(v, w)$ defined on L_2 space, we say that $\mathcal{H}$ satisfies the IQC defined by σ_{Π} , or “ $\mathcal{H}$ satisfies the IQC defined by Π ” to emphasize the multiplier involved, if $\sigma_{\Pi}(v, \mathcal{H}(v)) \geq 0$ for all $v \in L_2$. The main stability criterion, the so-called IQC theorem, given in (Megretski & Rantzer, 1997) is formulated as follows.

Theorem 1 (Megretski & Rantzer, 1997). *Let $G(s) \in \mathbf{RH}_{\infty}^{l \times m}$ and let Δ be a bounded causal operator. Suppose*

- (1) *for every $\rho \in [0, 1]$, the interconnection of G and $\rho\Delta$ is well-posed;*
- (2) *for every $\rho \in [0, 1]$, the IQC defined by Π is satisfied by $\rho\Delta$;*
- (3) *there exists $\varepsilon < 0$ such that*

$$\begin{bmatrix} G(j\omega) \\ I \end{bmatrix}^* \Pi(j\omega) \begin{bmatrix} G(j\omega) \\ I \end{bmatrix} \leq \varepsilon I, \quad \forall \omega \in [0, \infty]. \quad (4)$$

Then the feedback interconnection of G and Δ is stable.

Notice that if $\Pi_{11} \geq 0$ and $\Pi_{22} \leq 0$, then Δ satisfies the IQC defined by Π implies that $\rho\Delta$ satisfies the same IQC for all $\rho \in [0, 1]$.

It is important to note that if $\rho\Delta$ satisfies IQCs defined by Π_i , $i = 1, \dots, n$, then the conic combination $x_1\Pi_1 + \dots + x_n\Pi_n$, $x_i \geq 0$ also defines an IQC for $\rho\Delta$. Hence, a sufficient condition for stability is the existence of $x_1, \dots, x_n \geq 0$ such that (4) holds for $\Pi := x_1\Pi_1 + \dots + x_n\Pi_n$.

Furthermore, assume that the overall Δ is diagonally structured by n sub-operators Δ_i , $i = 1, \dots, n$; i.e., $\Delta = \text{diag}(\Delta_1, \dots, \Delta_n)$ and each Δ_i satisfies an IQC defined by Π_i . Then, an IQC for Δ can be easily defined by composing Π_i

appropriately. Hence, the most important feature of the IQC analysis is that it allows one to localize the analysis of complex dynamical systems to their elementary building blocks. In essence, the main task is to discover/derive IQCs for the non-LTI subsystems that contribute to the Δ operator. Once the IQCs for these subsystems are available, stability analysis for the overall system is then a rather straightforward matter of finding a single aggregate IQC for the overall Δ such that condition (4) holds.

Condition (4) is a frequency-dependent, infinite-dimensional linear matrix inequality (LMI). Suppose that $\Pi \in \mathbf{RL}_{\infty}$. Then this matrix inequality can be converted into a frequency-independent finite-dimensional LMI using the Kalman–Yakubovich–Popov (KYP) lemma. See Megretski and Rantzer (1997) for details.

3. IQC for $\mathcal{D}_{\tau}$ and $\mathcal{S}_{\tau}$

In this section, conically parameterized IQC for operators $\mathcal{D}_{\tau}$ and $\mathcal{S}_{\tau}$ are derived. These IQCs will be used in the next section to derive stability criteria for linear systems with time-varying delays.

Lemma 1. *Suppose that $d < 1$. Consider the set of differentiable functions $\mathcal{Y} := \{s(t) | s(t) \in [0, \tau_0], |\dot{s}(t)| \leq d, \forall t\}$. Then the following equalities hold:*

$$\sup_{\tau(t) \in \mathcal{Y}} \|\mathcal{D}_{\tau}\|_{L_2} = \frac{1}{\sqrt{1-d}}, \quad \sup_{\tau(t) \in \mathcal{Y}} \left\| \mathcal{S}_{\tau} \circ \frac{1}{s} \right\|_{L_2} = \tau_0.$$

Proof. The proof that $\|\mathcal{D}_{\tau}\|_{L_2}$ and $\|\mathcal{S}_{\tau} \circ 1/s\|_{L_2}$ are bounded by $1/\sqrt{1-d}$ and τ_0 , respectively, can be found in, for example, Gu et al. (2003) and Kao and Lincoln (2004). To see that $\sup_{\tau(t) \in \mathcal{Y}} \|\mathcal{D}_{\tau}\|_{L_2} = 1/\sqrt{1-d}$, let us consider the following signals:

$$v(t) = \begin{cases} 1, & t \in \left[0, \frac{\tau_0}{d} - \tau_0\right], \\ 0 & \text{otherwise,} \end{cases} \quad \tau(t) = \begin{cases} dt, & t \in \left[0, \frac{\tau_0}{d}\right], \\ \tau_0 & \text{otherwise.} \end{cases}$$

Then one can easily verify that $v(t - \tau(t))$ is equal to 1 for $t \in [0, \tau_0/d]$, and equal to 0 otherwise. Hence we have

$$\frac{\|v(t - \tau(t))\|_{L_2}^2}{\|v(t)\|_{L_2}^2} = \frac{1}{1-d}.$$

To see that $\sup_{\tau(t) \in \mathcal{Y}} \|\mathcal{S}_{\tau} \circ 1/s\|_{L_2} = \tau_0$, let $\tau(t) = \tau_0$ for all t . Then

$\mathcal{S}_{\tau}$ is a linear time-invariant operator. The transfer function of $\mathcal{S}_{\tau}$ is equal to $\Delta(s) \cdot I$, where $\Delta(s) = (e^{-s\tau_0} - 1)/s$. The L_2 -induced norm of $\Delta(s) \cdot I$ is equal to $\sup_{\omega} |\Delta(j\omega)|$, which is equal to τ_0 . This concludes the proof. $\square$

Remark 1. It is interesting to note that as long as $\tau_0 > 0$, the bound on $\|\mathcal{D}_{\tau}\|_{L_2}$ depends on the variation of τ (i.e., d) but does not depend on its length (i.e., τ_0). In contrast, $\|\mathcal{S}_{\tau} \circ 1/s\|_{L_2}$ depends on τ_0 but does not depend on d .

Lemma 2. Suppose that $0 \leq \tau(t) \leq \tau_0$ and $|\dot{\tau}(t)| \leq d < 1$ for all t . Then for any $v \in \mathcal{L}_2$ such that $\text{Supp}(|\hat{v}(j\omega)|) = [\omega_1, \omega_2]$ and $|\omega_2 - \omega_1| := \varepsilon$ is sufficiently small, we have

$$\|\mathcal{S}_\tau(v)\|_{L_2} \leq \left(\sup_{\omega \in [\omega_1, \omega_2]} g(\omega) + \mathbf{O}(\varepsilon) \right) \|v\|_{L_2},$$

where the function $g(\omega)$ is defined as follows:

$$g(\omega) = \begin{cases} \sqrt{\frac{8}{2-d}}, & |\omega| > \frac{\pi}{\tau_0}, \\ \sqrt{\frac{8}{2-d}} \left| \sin\left(\frac{\omega\tau_0}{2}\right) \right|, & |\omega| \leq \frac{\pi}{\tau_0}. \end{cases} \quad (5)$$

Proof. To streamline the readability of the paper, the proof of Lemma 2 is placed in the Appendix.

The bounds on $\|\mathcal{D}_\tau\|_{L_2}$ and $\|\mathcal{S}_\tau\|_{L_2}$ give rise to the following IQC for $\mathcal{D}_\tau$ and $\mathcal{S}_\tau$.

Proposition 1. Suppose $|\dot{\tau}| \leq d < 1$. Then the operator $\mathcal{D}_\tau$ satisfies any IQC defined by

$$\Pi_1 = \begin{bmatrix} \frac{1}{1-d} X_1 & 0 \\ 0 & -X_1 \end{bmatrix}, \quad (6)$$

where $X_1 = X'_1 \geq 0$ is any positive semi-definite matrix.

Proof. Proposition 1 follows Lemma 1 and that, given a positive semi-definite matrix X_1 , $(X_1^{1/2} \mathcal{D}_\tau)v = (\mathcal{D}_\tau X_1^{1/2})v$. $\square$

Proposition 2. Suppose $\tau(t) \in [0, \tau_0]$ and $|\dot{\tau}(t)| \leq d < 1$ for all t . Then the operator $\mathcal{S}_\tau$ satisfies any IQC defined by

$$\Pi_2 = \begin{bmatrix} |\phi(j\omega)|^2 X_2 & 0 \\ 0 & -X_2 \end{bmatrix}, \quad (7)$$

where $\phi(s)$ is any bounded rational transfer function which satisfies

$$\begin{cases} |\phi(j\omega)| > 1 + \frac{1}{\sqrt{1-d}} & \text{if } \tau_0|\omega| > 1 + \frac{1}{\sqrt{1-d}}, \\ |\phi(j\omega)| > \tau_0|\omega| & \text{if } \tau_0|\omega| \leq 1 + \frac{1}{\sqrt{1-d}}, \end{cases} \quad (8)$$

and $X_2 = X'_2 \geq 0$ is any positive semi-definite matrix.

Remark 2. An example of $\phi(s)$ satisfying (8) is

$$k \cdot \frac{\tau_0^2 s^2 + c\tau_0 s}{\tau_0^2 s^2 + a\tau_0 s + kc},$$

where $k = 1 + 1/\sqrt{1-d}$, $a = \sqrt{2kc}$, and c is any positive real number.

Proof. The key idea behind the IQC is that the L_2 -induced norm of operator $\mathcal{S}_\tau \circ \phi^{-1}$ is less than or equal to 1. With this fact, the IQC follows that, given any $v \in L_2$ and positive semi-definite matrix X_2 ,

$$X_2^{1/2} (\mathcal{S}_\tau \circ \phi^{-1})v = (\mathcal{S}_\tau \circ \phi^{-1})X_2^{1/2}v.$$

To show $\|\mathcal{S}_\tau \circ \phi^{-1}\|_{L_2} \leq 1$, let v be any L_2 function. Furthermore, decompose v into $v_1 + v_2$ such that the Fourier transforms of v_1 and v_2 satisfy

$$\text{Supp}(|\hat{v}_1(j\omega)|) = \Omega_1, \quad \text{Supp}(|\hat{v}_2(j\omega)|) = \Omega_2,$$

where

$$\Omega_1 := \left\{ \omega : \tau_0|\omega| > 1 + \frac{1}{\sqrt{1-d}} \right\},$$

$$\Omega_2 := \left\{ \omega : \tau_0|\omega| \leq 1 + \frac{1}{\sqrt{1-d}} \right\}.$$

Since $\mathcal{S}_\tau \circ \phi^{-1}$ is a linear operator,

$$\begin{aligned} (\mathcal{S}_\tau \circ \phi^{-1})(v_1 + v_2) &= (\mathcal{S}_\tau \circ \phi^{-1})v_1 + (\mathcal{S}_\tau \circ \phi^{-1})v_2 \\ &= \left(\mathcal{S}_\tau \left(1 + \frac{1}{\sqrt{1-d}} \right)^{-1} \left(1 + \frac{1}{\sqrt{1-d}} \right) \phi^{-1} \right) v_1 \\ &\quad + \left(\mathcal{S}_\tau \circ \frac{1}{\tau_0 s} \circ (\tau_0 s) \circ \phi^{-1} \right) v_2. \end{aligned}$$

Note that $\|(\mathcal{S}_\tau \circ \phi^{-1})v_1\|_{L_2}$ is bounded by

$$\begin{aligned} &\left\| \mathcal{S}_\tau \left(1 + \frac{1}{\sqrt{1-d}} \right)^{-1} \right\|_{L_2} \cdot \left\| \left(1 + \frac{1}{\sqrt{1-d}} \right) \phi^{-1} v_1 \right\|_{L_2} \\ &\leq \left\| \left(1 + \frac{1}{\sqrt{1-d}} \right) \phi^{-1} v_1 \right\|_{L_2} \leq \|v_1\|_{L_2}. \end{aligned}$$

The first inequality follows that $\|\mathcal{S}_\tau\|_{L_2} \leq 1 + \|\mathcal{D}_\tau\|_{L_2}$. The second inequality follows that $\text{Supp}(|\hat{v}_1(j\omega)|) = \Omega_1$ and that $|\phi(j\omega)| > 1 + 1/\sqrt{1-d}$ for all $\omega \in \Omega_1$. Similarly,

$$\begin{aligned} &\left\| \left(\mathcal{S}_\tau \circ \frac{1}{\tau_0 s} \circ (\tau_0 s) \circ \phi^{-1} \right) v_2 \right\|_{L_2} \\ &\leq \left\| \mathcal{S}_\tau \circ \frac{1}{\tau_0 s} \right\|_{L_2} \cdot \|(\tau_0 s) \circ \phi^{-1} v_2\|_{L_2} \leq \|v_2\|_{L_2}. \end{aligned}$$

Therefore,

$$\begin{aligned} \|\mathcal{S}_\tau \circ \phi^{-1}v\|_{L_2} &\leq \|\mathcal{S}_\tau \circ \phi^{-1}v_1\|_{L_2} + \|\mathcal{S}_\tau \circ \phi^{-1}v_2\|_{L_2} \\ &\leq \|v_1\|_{L_2} + \|v_2\|_{L_2} = \|v_1 + v_2\|_{L_2} = \|v\|_{L_2}. \end{aligned}$$

This concludes the proof. $\square$

Proposition 3. Suppose $\tau(t) \in [0, \tau_0]$ and $|\dot{\tau}(t)| \leq d < 2$ for all t . Then the operator $\mathcal{S}_\tau$ satisfies any IQC defined by

$$\Pi_3 = \begin{bmatrix} |\psi(j\omega)|^2 X_3 & 0 \\ 0 & -X_3 \end{bmatrix}, \quad (9)$$

where $X_3 = X'_3 \geq 0$ is any positive semi-definite matrix, and $\psi(s)$ is any bounded rational transfer function which satisfies

$$|\psi(j\omega)| \geq g(\omega) + \delta, \quad \forall \omega \in \mathbb{R}. \quad (10)$$

Here, function $g(\omega)$ is defined as in (5) and δ is some positive real number.

Remark 3. An example of $\psi(s)$ which satisfies (10) is

$$k \cdot \frac{\tau_0^2 s^2 + c\tau_0 s}{\tau_0^2 s^2 + a\tau_0 s + b} + \delta,$$

where $k = \sqrt{8/(2-d)}$, $a = \sqrt{6.5 + 2b}$, $b = \sqrt{50}$, $c = \sqrt{12.5}$, and δ is an arbitrarily small positive number.

Proof. The proof of Proposition 3 is analog to that of Proposition 2. The key idea behind the IQC is that $\|\mathcal{S}_\tau \circ \psi^{-1}\|_{L_2} \leq 1$. To show this, first note that $g(\omega)$ is uniformly continuous on $\mathbb{R}$. Since $|\psi(j\omega)| - g(\omega) \geq \delta$ for all $\omega \in \mathbb{R}$ and $g(\omega)$ is uniformly continuous, there exists an ε^* such that any interval A which satisfies $|A| \leq \varepsilon^*$, the following inequality holds:

$$\inf_{\omega \in A} |\psi(j\omega)| \geq \sup_{\omega \in A} g(\omega) + \frac{\delta}{2}. \quad (11)$$

Now, consider a partition of $\mathbb{R}$, $\{A_k\}$, such that each A_k satisfies $|A_k| := \varepsilon \leq \varepsilon^*$ for all k . For any given $v(t) \in L_2$, let $v(t) = \sum_k v_k(t)$ such that

$$\hat{v}_k(j\omega) = \begin{cases} \hat{v}(j\omega), & \omega \in A_k, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\mathcal{S}_\tau \circ \psi^{-1}$ is a linear operator, we have

$$(\mathcal{S}_\tau \circ \psi^{-1})v = \sum_k (\mathcal{S}_\tau \circ \psi^{-1})v_k.$$

Now let $\gamma_k := \sup_{\omega \in A_k} g(\omega) + \delta/2$. Note that the signal $\gamma_k \psi^{-1}v_k$ is non-zero only on A_k . Therefore,

$$\begin{aligned} \|(\mathcal{S}_\tau \circ \psi^{-1})v_k\|_{L_2} &= \|(\gamma_k^{-1} \mathcal{S}_\tau) \circ (\gamma_k \psi^{-1})v_k\|_{L_2} \\ &\leq \gamma_k^{-1} \left(\sup_{\omega \in A_k} g(\omega) + \mathbf{O}(\varepsilon) \right) \|(\gamma_k \psi^{-1})v_k\|_{L_2} \\ &\leq \|v_k\|_{L_2}. \end{aligned}$$

The first inequality follows Lemma 2. To see the second inequality, note that $\gamma_k > \sup_{\omega \in A_k} g(\omega) + \mathbf{O}(\varepsilon)$ for sufficiently small ε , and that

$$\begin{aligned} \|(\gamma_k \psi^{-1})v_k\|_{L_2}^2 &= \int_{\omega \in A_k} \gamma_k^2 |\psi(j\omega)|^{-2} \|V_k(j\omega)\|^2 d\omega \\ &\leq \int_{\omega \in A_k} \|V_k(j\omega)\|^2 d\omega \\ &= \|v_k\|_{L_2}^2. \end{aligned}$$

Finally,

$$\begin{aligned} \|\mathcal{S}_\tau \circ \psi^{-1}v\|_{L_2} &= \left\| \sum_k (\mathcal{S}_\tau \circ \psi^{-1})v_k \right\|_{L_2} \\ &\leq \sum_k \|(\mathcal{S}_\tau \circ \psi^{-1})v_k\|_{L_2} \\ &\leq \sum_k \|v_k\|_{L_2} = \|v\|_{L_2}. \end{aligned}$$

This concludes the proof. $\square$

The IQCs derived in Propositions 1–3 are d -dependent, where d must be less than either 1 or 2. Based on $\|\mathcal{S}_\tau \circ 1/s\|_{L_2} \leq \tau_0$, the following d -independent IQC can be derived for $\mathcal{S}_\tau$.

Proposition 4. $\mathcal{S}_\tau$ satisfies the IQC defined by

$$\Pi_4 = \begin{bmatrix} \tau_0^2 \omega^2 X_4 & 0 \\ 0 & -X_4 \end{bmatrix}, \quad (12)$$

where $X_4 = X_4' \geq 0$.

Proof. A direct consequence of $\|\mathcal{S}_\tau \circ 1/s\|_{L_2} \leq \tau_0$. $\square$

Remark 4. Since Π_4 is not a bounded operator on L_2 , the quadratic form defined by Π_4 is not defined on the entire L_2 space. Therefore, $\mathcal{S}_\tau$ satisfies the IQC only when the input signal has an L_2 derivative. Π_4 may be used in IQC analysis as long as the ω^2 term can be absorbed into the nominal system G ; i.e., G must be strictly proper.

The scaling matrices X_1 , X_2 , X_3 , and X_4 of IQCs defined in (6), (7), (9), and (12) are constant (as opposed to frequency-dependent) over all frequencies. Frequency-dependent scalings, which usually provide better characterization of $\mathcal{D}_\tau$ and $\mathcal{S}_\tau$, are allowed if certain correction terms are added to the IQCs. The key idea behind is the following swapping lemma.

Lemma 3 (swapping lemma). Let $H(s) := C(sI - A)^{-1}B + D$, $H_l(s) := C(sI - A)^{-1}$, and $H_r(s) := (sI - A)^{-1}B$ be proper rational transfer matrices from $\mathbf{RL}_\infty^{n \times n}$. Furthermore, let T denote the operator of multiplying $\dot{\tau}$; i.e., $T(v(t)) := \dot{\tau} \cdot v(t)$. Then,

$$\mathcal{D}_\tau \circ H(s) = H(s) \circ \mathcal{D}_\tau - H_l(s) \circ T \circ \mathcal{D}_\tau \circ s H_r(s). \quad (13)$$

Proof. Let $y = \mathcal{D}_\tau \circ H v$ and $z = H \circ \mathcal{D}_\tau(v)$. Then, $y(t) = Cx_1(t - \tau) + Dv(t - \tau)$, where x_1 is the state of the system

$$\dot{x}_1(t) = Ax_1(t) + Bv(t), \quad x_1(0) = 0,$$

and $z(t) = Cx_2(t) + Dv(t - \tau)$, where x_2 satisfies

$$\dot{x}_2(t) = Ax_2(t) + Bv(t - \tau), \quad x_2(0) = 0.$$

Let $w = y - z = C(x_1(t - \tau) - x_2(t)) = Cx_3(t)$. Differentiating x_3 w.r.t. time, we obtain

$$\begin{aligned} \dot{x}_3(t) &= \dot{x}_1(t - \tau) - \dot{x}_2(t) - \dot{\tau} \cdot \dot{x}_1(t - \tau) \\ &= Ax_3(t) - \dot{\tau} \cdot \dot{x}_1(t - \tau). \end{aligned}$$

Hence, we have shown that

$$\begin{aligned} w &:= (\mathcal{D}_\tau \circ H - H \circ \mathcal{D}_\tau)v = H_l(s)(-\dot{\tau} \cdot \mathcal{D}_\tau(\dot{x}_1)) \\ &= -H_l(s) \circ T \circ \mathcal{D}_\tau \circ s H_r(s)v. \end{aligned}$$

This concludes the proof. $\square$

Remark 5. Using (13), the following equalities can be readily verified:

$$\begin{aligned} (\mathcal{S}_\tau \circ \phi^{-1}) \circ H(s) &- H(s) \circ (\mathcal{S}_\tau \circ \phi^{-1}) \\ &= H_l(s) \circ T \circ \mathcal{D}_\tau \circ s H_r(s) \phi^{-1}, \end{aligned}$$

$$(\mathcal{S}_\tau \circ \psi^{-1}) \circ H(s) - H(s) \circ (\mathcal{S}_\tau \circ \psi^{-1}) \\ = H_l(s) \circ T \circ \mathcal{D}_\tau \circ s H_r(s) \psi^{-1},$$

$$(\mathcal{S}_\tau \circ s^{-1}) \circ H(s) - H(s) \circ (\mathcal{S}_\tau \circ s^{-1}) \\ = H_l(s) \circ T \circ \mathcal{D}_\tau \circ s H_r(s) s^{-1},$$

where $\phi(s)$ and $\psi(s)$ are rational transfer functions defined in (8) and (10), respectively.

The swapping formulas lead to the following IQC characterization for $\mathcal{D}_\tau$ and $\mathcal{S}_\tau$.

Proposition 5. Let $H(s) := h(s) \cdot I_n$, where $h(s) \in \mathbf{RL}_\infty^{1 \times 1}$, and the corresponding $H_l(s)$ and $H_r(s)$ be defined similarly as those in Lemma 3. Suppose $\tau \in [0, \tau_0]$ and $|\dot{\tau}| \leq d < 1$. Then, $\mathcal{D}_\tau$ satisfies the IQC defined by Π_5 , and $\mathcal{S}_\tau$ satisfies IQC defined by Π_6 , Π_7 , and Π_8 , where Π_5 , Π_6 , Π_7 , and Π_8 are of the following forms:

$$\Pi_5 = \begin{bmatrix} \frac{1}{1-d} H^* X_5 H & 0 \\ 0 & -H^* X_5 H \end{bmatrix} + \Pi_{C,5}, \quad (14)$$

$$\Pi_6 = \begin{bmatrix} |\phi(j\omega)|^2 H^* X_6 H & 0 \\ 0 & -H^* X_6 H \end{bmatrix} + \Pi_{C,6}, \quad (15)$$

$$\Pi_7 = \begin{bmatrix} |\psi(j\omega)|^2 H^* X_7 H & 0 \\ 0 & -H^* X_7 H \end{bmatrix} + \Pi_{C,7}, \quad (16)$$

$$\Pi_8 = \begin{bmatrix} \tau_0^2 \omega^2 H^* X_8 H & 0 \\ 0 & -H^* X_8 H \end{bmatrix} + \Pi_{C,8}, \quad (17)$$

where $\Pi_{C,i}$, $i = 5, 6, 7, 8$, are of the form

$$\begin{bmatrix} \frac{d}{1-d} \cdot \omega^2 H_r^* X_i H_r & 0 \\ 0 & d \cdot H^* H_l X_i H_l^* H \end{bmatrix},$$

and $X_i = X_i'$, $i = 5, 6, 7, 8$, are any positive semi-definite matrices.

Proof. We will show that operator $\mathcal{D}_\tau$ satisfies the IQC defined by Π_5 . The other IQCs can be derived in a similar fashion. Given a L_2 signal v , let $w = \mathcal{D}_\tau(v)$. Note that

$$\begin{aligned} \langle Hw, Hw \rangle &= \langle Hw, (\mathcal{D}_\tau \circ H)v + (H_l \circ T \circ \mathcal{D}_\tau \circ s H_r)v \rangle \\ &= \langle Hw, (\mathcal{D}_\tau \circ H)v \rangle \\ &\quad + \left\langle \sqrt{d} H_l^* Hw, \frac{1}{\sqrt{d}} (T \circ \mathcal{D}_\tau \circ s H_r)v \right\rangle. \end{aligned}$$

Hence,

$$\begin{aligned} 2\langle Hw, Hw \rangle &\leq \langle Hw, Hw \rangle + \frac{1}{1-d} \langle Hv, Hv \rangle \\ &\quad + d \langle H_l^* Hw, H_l^* Hw \rangle + \frac{d}{1-d} \langle s H_r v, s H_r v \rangle, \end{aligned}$$

which implies

$$\left\langle \begin{bmatrix} v \\ w \end{bmatrix}, \begin{bmatrix} M_1 & 0 \\ 0 & M_2 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} \right\rangle \geq 0,$$

where $M_1 = (1/(1-d))H^*H + (d/(1-d))(sH_r)^*(sH_r)$ and $M_2 = -H^*H + d(H_l^*H)^*H_l^*H$. Finally, to see the conic

parametrization of Π_5 , note that given any positive semi-definite matrix X , matrix $X^{1/2}$ commutes with $\mathcal{D}_\tau$, $H(s)$, $H_l(s)$, and $sH_r(s)$. This concludes the proof. $\square$

Remark 6. Note that the IQCs stated in Propositions 1–4 are special cases of those stated in Proposition 5. Indeed, by choosing $h(s) \equiv 1$, the multipliers defined in (14)–(17) reduce to those defined in (6), (7), (9), and (12), respectively.

4. Stability criteria for linear time-varying delay systems

Consider now the linear time-varying delay system

$$\dot{x} = Ax + A_d(x(t-\tau) + f), \quad (18)$$

where f is a L_2 disturbance. Delay parameter τ satisfies condition (2). $A + A_d$ is assumed to be Hurwitz, which is a necessary condition for stability.

The system can be modelled as the feedback interconnection

$$x = Gw + e, \quad w = \Delta(x), \quad (19)$$

where $G := (sI - (A + A_d))^{-1}A_d$ is a linear time-invariant stable system, $e = Gf$, and $\Delta := -\mathcal{S}_\tau$. With the IQC derived in Section 3, IQC analysis can be applied to the transformed system to study L_2 stability of system (18). Note that G is strictly proper; hence IQC defined by Π_4 is also valid to be used in the analysis.

4.1. Case: $d < 1$

When the upper bound of $\dot{\tau}(t)$ is strictly less than 1, IQCs stated in Propositions 1–5 are all valid. Following the IQC theorem presented in Section 2, these IQCs can be used to compose various L_2 stability criteria for system (18). Notice that any IQC for $\mathcal{D}_\tau$ immediately leads to an IQC for $\mathcal{S}_\tau$. For example, let $w = \mathcal{S}_\tau v := v - \mathcal{D}_\tau v$. That $\mathcal{D}_\tau$ satisfies IQC defined by Π_1 in (6) implies v and w satisfy IQC

$$\left\langle \begin{bmatrix} v \\ w \end{bmatrix}, \begin{bmatrix} \frac{d}{1-d} X_1 & X_1 \\ X_1 & -X_1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} \right\rangle \geq 0.$$

Stability criteria derived via IQC analysis are naturally posed as semi-infinite optimization problems. Finite dimensional formulation (in terms of linear matrix inequalities) can be derived using the KYP lemma.

4.2. Case: $1 \leq d < 2$

When $1 \leq d < 2$, the L_2 gain of $\mathcal{D}_\tau$ is infinite. As a result, among those IQCs stated in Propositions 1–5, only those defined by Π_3 and Π_4 in (9) and (12) are applicable to characterize $\mathcal{S}_\tau$. Stability criteria in this case can be derived via IQC analysis with the IQC defined by multipliers Π_3 and Π_4 .

Remark 7. Essentially, the case $d > 1$ corresponds to the scenario where we use a measurement (say, $y(t_1)$) that is older than the ones (say, $y(t_2)$) we have already got and used before. That is, we use $y(t_2)$ at time instance t_3 and $y(t_1)$ at time

instance t_4 where $t_4 > t_3 > t_2 > t_1$. Such scenario might be introduced intentionally, for the “older” measurement still carries some information that is crucial. Or this might happen because the signals are sent without a time stamp, or the time stamp is lost or corrupted.

4.3. Case: $d \geq 2$ or upper bound on $\dot{\tau}(t)$ is not available

In this case, only the IQC defined by Π_4 in (12) is applicable. Using this IQC in the IQC analysis corresponds to applying the scaled small gain theorem to the following system:

$$\dot{x} = (A + A_d)x - A_d w + A_d f,$$

$$\begin{aligned} w &:= \int_{t-\tau}^t \dot{x}(\theta) d\theta \\ &:= \left(\mathcal{S}_\tau \circ \frac{1}{s} \right) ((A + A_d)x - A_d w + A_d f). \end{aligned} \quad (20)$$

It is well known that the *descriptor* model (20) is *not* equivalent to system (1). The transformation introduces additional dynamics. The stability condition for the descriptor model (20) must also take into account the additional dynamics. This makes the condition conservative. For the case where τ is constant, the additional dynamics is characterized in Gu and Niculescu (2000), where it is shown that the degree of conservatism has to do with the location of the eigenvalues of A_d .

However, stability of system (20) does imply stability of system (18); hence the condition is sufficient. The delay-derivative-independent stability criterion is stated in the following proposition.

Proposition 6. Suppose $A + A_d$ has no eigenvalue in the closed right half plane. Then system (20) is stable if there exists a symmetric positive semi-definite matrix X of suitable dimension, such that for some $\varepsilon > 0$,

$$\tau_0^2 G(j\omega)^* X G(j\omega) - X \leq -\varepsilon I, \quad \forall \omega \in [0, \infty), \quad (21)$$

where $G(s) = (A + A_d)(sI - (A + A_d))^{-1} A_d + A_d$.

By the KYP lemma, stability criterion (21) can be expressed in terms of the LMI

$$\begin{aligned} &\begin{bmatrix} P(A + A_d) + (A + A_d)'P & P A_d \\ A_d' P & 0 \end{bmatrix} \\ &+ \begin{bmatrix} \tau_0^2 (A + A_d)' X (A + A_d) & \tau_0^2 (A + A_d)' X A_d \\ \tau_0^2 A_d' X (A + A_d) & \tau_0^2 A_d' X A_d - X \end{bmatrix} \leq 0, \end{aligned} \quad (22)$$

where $P = P'$ is a symmetric matrix.

The idea presented here is not new; a very similar result can be found in Gu et al. (2003). Furthermore, many stability criteria developed under the Lyapunov framework also have the small-gain IQC for the operator $\mathcal{S}_\tau \circ 1/s$ embedded in the Lyapunov–Krasovskii functionals which lead to conditions similar or identical to LMI (22).

5. Remark on the extension to systems with multiple delays and other uncertainty and nonlinearity

Consider now the following dynamical system:

$$\begin{aligned} \dot{x}(t) &= Ax(t) + \sum_{i=1}^n A_{d,i} x(t - \tau_i(t)) \\ &\quad + \Delta_p(x(t), x(t - \tau_1), \dots, x(t - \tau_n)) + f, \end{aligned} \quad (23)$$

where Δ_p is a bounded and causal operator which may be uncertain, time-varying, and/or nonlinear. Signal f is a L_2 disturbance. We again assume that each delay parameter τ_i satisfies condition (2) and that $A + \sum_{i=1}^n A_{d,i}$ is a Hurwitz matrix.

Through an appropriate loop transformation, the system can be modelled as a feedback interconnection

$$v = G(s)w + e, \quad w = \Delta(v), \quad (24)$$

where $G(s)$ is a stable linear time-invariant system, e is an external disturbance arising from f , $\Delta = \text{diag}(\Delta_\tau, \Delta_p)$, and

$$\Delta_\tau = \begin{bmatrix} \mathcal{S}_{\tau_1} \\ \vdots \\ \mathcal{S}_{\tau_n} \end{bmatrix}.$$

Suppose that an IQC characterization of Δ_p is available. Then, IQCs stated in Propositions 1–5 can be used, together with the IQC which characterizes Δ_p , to compose various L_2 stability criteria for system (23). Assembling the corresponding IQCs can be automated by a custom-made MATLAB toolbox. Readers interested in more details about this are referred to Kao, Megretski, Jönsson, and Rantzer (2004).

6. Numerical examples and comparison

In this section, numerical experiments are presented to demonstrate the IQC analysis proposed in this paper.

Example 1. Consider the following system (Li & de Souza, 1997a, Example 4),

$$\begin{aligned} \dot{x}(t) &:= Ax(t) + A_d x(t - \tau) \\ &= \begin{bmatrix} -2 & 0 \\ 0 & -0.9 \end{bmatrix} x(t) + \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix} x(t - \tau). \end{aligned} \quad (25)$$

If τ is a constant, the Nyquist criterion shows that when $\tau = \arccos(-0.9)/\sqrt{0.19} \approx 6.1725$, the system has an eigenvalue on the imaginary axis.

We use the following IQCs in our analysis:

- $\dot{\tau} \leq 0.8$: the IQCs defined by Π_1 and Π_3 ,
- $0.8 < \dot{\tau} \leq 0.999$: the IQCs defined by Π_1 , Π_3 , and Π_4 ,
- $1 \leq \dot{\tau} \leq 1.999$: the IQCs defined by Π_3 and Π_4 ,
- $\dot{\tau} \geq 2$: the IQC defined by Π_4 .

To demonstrate the effectiveness, results are compared against those obtained by using criteria in Kim (2001), Fridman and Shaked (2002) and Wu et al. (2004). The results are shown in

Table 1
The allowable time delays (Example 1)

d	0	0.1	0.2	0.5	0.8	0.999	1	1.25	1.5	1.75	1.999	≥ 2
Kim (2001)	1	0.974	0.883	0.655	0.322	0.001	—	—	—	—	—	—
Wu et al. (2004)	4.472	3.604	3.033	2.008	1.364	1.001	—	—	—	—	—	—
Fridman and Shaked (2002)	4.472	3.604	3.033	2.008	1.364	1.001	0.999	0.999	0.999	0.999	0.999	0.999
IQC analysis	6.117	4.714	3.807	2.280	1.608	1.360	1.359	1.208	1.119	1.054	1.000	0.999

Table 2
Decision variables need to be optimized (Example 1)

Kim	Wu et al.	Fridman and Shaked (2002)	IQC (Π_1, Π_3, Π_4)	IQC (Π_1, Π_3)	IQC (Π_3, Π_4)	IQC (Π_4)
9	27	35	31	28	28	7

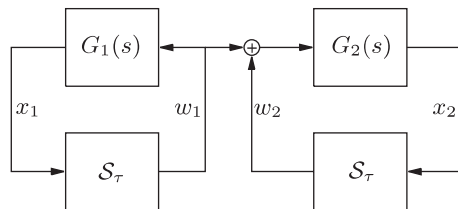


Fig. 1. The cascade delay system.

Table 1. The decision variables to be optimized in the analysis are compared in **Table 2**. We observe that in this example IQC analysis produces the least conservative results among all methods. Furthermore, when $d = 0$, the IQC analysis almost recovers the true stability margin (6.1725). The accuracy is 99%. Some explanation on how this result is achieved is given below.

The system (25) is a cascade of two subsystems as illustrated in **Fig. 1**, where $G_1(s) = 1/(s + 3)$ and $G_2(s) = 1/(s + 1.9)$. The left-hand side subsystem is delay-independently stable if $\dot{\tau} < 0.5$, and therefore can be ignored in such cases. In such cases, we have verified our analysis and confirmed that delay bounds obtained by considering only the right-hand side subsystem are *identical to* (less a minor 10^{-4} numerical inaccuracy) those obtained for the whole systems. In fact, the bounds are identical in all cases, which is expected because $G_1(s)$ is “more stable” than $G_2(s)$ from energy amplification point of view.

In our analysis, when $\dot{\tau} \leq 0.8$, the stability criterion for the right-hand side system turns out to be

$$X_1(|G_2(j\omega)\psi(j\omega)|^2 - 1) + X_2(2\operatorname{Re} G_2(j\omega) - 1) < 0, \quad \forall \omega \in [0, \infty],$$

where $\psi(s)$ is given in Remark 3, and X_1, X_2 are two positive numbers. Consider the case where τ is time-invariant (i.e., $d = 0$). In this case, the above condition is almost tight. Indeed, the term with X_2 indicates that the critical frequency range is $\omega \in [0, \sqrt{0.19}]$. Outside this range, $2\operatorname{Re} G_2(j\omega) < 1$ and the inequality is satisfied by letting $X_2 \gg X_1 > 0$. It can be easily verified (by viewing the right-hand side system as $1/(s + 0.9)$

in feedback interconnection with a delay) that if a disturbance input to the system in **Fig. 1** only has frequency components in the range $(\sqrt{0.19}, \infty)$, all signals in the loop will remain bounded. In the frequency range $[0, \sqrt{0.19}]$, what IQC analysis does is replacing the “delay difference” operator $1 - e^{-s\tau}$ by $\psi(s)$, and deducing stability by the small gain argument. Note that for the operator $1 - e^{-s\tau}$ (where $0 \leq \tau \leq \tau_0$), the worst case L_2 -gain occurs when $\tau = \tau_0$. The multiplier $\psi(s)$ is constructed in such a way that $|\psi(j\omega)|$ is larger than but very close to $|1 - e^{-j\omega\tau_0}|$. The 1% conservativeness is precisely due to the difference between $|\psi(j\omega)|$ and $|1 - e^{-j\omega\tau_0}|$ in the frequency range of $[0, \sqrt{0.19}]$.

For time-varying τ , the IQC analysis using multipliers Π_1 and Π_3 can be understood exactly like the above stated. However, it is difficult to grasp the degree of conservativeness because we do not know how well $\psi(s)$ approximates $1 - \mathcal{D}_\tau$ from energy amplification point of view.

Example 2. Consider the following system (Kim, 2001, Example 1):

$$\dot{x}(t) = \begin{bmatrix} -2 + a_1(t) & 0 \\ 0 & -1 + a_2(t) \end{bmatrix} x(t) + \begin{bmatrix} -1 + a_3(t) & 0 \\ -1 & -1 + a_4(t) \end{bmatrix} x(t - \tau(t)), \quad (26)$$

where τ is a time-varying parameter. Coefficients $a_i(t)$ are harmonic functions with bounded unknown amplitude: $a_1(t) = \delta_1 \cos(t)$, $a_2(t) = \delta_2 \sin(t)$, $a_3(t) = \gamma_1 \cos(t)$, $a_4(t) = \gamma_2 \sin(t)$, where $|\delta_1| \leq 1.6$, $|\delta_2| \leq 0.05$, $|\gamma_1| \leq 0.1$, $|\gamma_2| \leq 0.3$.

In this example, we have two sets of IQC tests. In the first set, we use the following IQCs for the “delay difference” operator $\mathcal{S}_\tau$:

- $\dot{\tau} \leq 0.999$: the IQCs defined by Π_1, Π_2 , and Π_3 ,
- $1 \leq \dot{\tau} \leq 1.999$: the IQCs defined by Π_3 and Π_4 ,
- $\dot{\tau} \geq 2$: the IQC defined by Π_4 ,

and simple “gain-bounded” IQCs for the time-varying coefficients $a_i(t)$. In the second set of IQC tests, we use the same

Table 3
The allowable time delays (Example 2)

d	0	0.1	0.2	0.5	0.8	0.999	1	1.25	1.5	1.75	1.99	≥ 2
Kim (2001)	0.241	0.234	0.225	0.188	0.110	0.0007	—	—	—	—	—	—
Wu et al. (2004)	1.149	1.106	1.063	0.924	0.760	0.628	—	—	—	—	—	—
IQC test 1	1.239	1.171	1.105	0.932	0.817	0.792	0.792	0.778	0.775	0.775	0.768	0.768
IQC test 2	1.416	1.302	1.201	0.974	0.829	0.796	0.796	0.779	0.776	0.776	0.768	0.768

Table 4
Decision variables need to be optimized (Example 2)

Kim	Wu et al.	IQC test 1	IQC test 2
15	27	Maximum 69	Maximum 275

IQCs as those used in the first set for $\mathcal{S}_\tau$, but change the IQCs for $a_i(t)$ to characterize their periodic nature (see Megretski & Rantzer, 1997 for details). The IQC analysis are compared against Kim (2001) and Wu et al. (2004). The results are shown in Table 3, while the decision variables to be optimized are given in Table 4.

From Table 3, we observe that the IQC tests give significantly better stability margins than those of Kim (2001). Furthermore, compared to Wu et al. (2004), the IQC test 1 in average improves the stability margins by 7% while IQC test 2 improves the margins by 13%. The price for such improvement is the computational complexity of the optimization problem to be solved: more than 200 decision variables need to be optimized in the IQC test 2, which is more than 3 times of those in the IQC test 1, 10 times of those in Wu et al. (2004), and 15 times of those in Kim (2001).

Another point we wish to make through this example is the adaptability of the IQC analysis. In the IQC test 2, more sophisticated IQCs are used for the uncertain time-varying coefficient $a_i(t)$, which significantly improves the stability margins when d is small. Note that, although deriving these IQCs may not be a trivial matter, once they are available, using them is as simple as changing a line of codes in the program routines. To improve the accuracy of the analysis, one shall only concentrate on improving the IQC characterization of individual components in the system.

7. Conclusions

Stability analysis in the IQC framework for linear time-varying delay systems are proposed. The delay parameter is assumed to be an unknown time-varying function for which the upper bounds on the magnitude and the variation are given. The influence of the time-varying delay is modelled as an uncertainty in the system, and IQC are derived to characterize the effect of the uncertain operator. Conditions for stability are then derived based on the IQC analysis. The advantage of this approach is that it can be easily generalized to accommodate systems with multiple delays, parametric uncertainties, unmodelled dynamics, and/or various simple nonlinearities.

Appendix A. Proof of Lemma 2

The main idea of the proof can be summarized as follows: given a L_2 function v such that the $\text{Supp}(|\hat{v}(j\omega)|) = [\omega_m, \omega_M]$, v can be approximated in norm arbitrarily well by a function $\underline{v}$ whose Fourier transform is a piecewise constant function. In particular, let $\omega_m := \omega_1 \leq \omega_2 \leq \dots \leq \omega_n := \omega_M$ is a partition of $[\omega_m, \omega_M]$ such that $\omega_{k+1} - \omega_k \leq \varepsilon$ for $k = 1, \dots, n$. An approximation of v may be given as

$$\underline{v}(t) = \sum_k \underline{v}_k(t),$$

where $\underline{v}_k(t)$ is defined in such a way that its Fourier transform satisfies

$$\hat{\underline{v}}_k(j\omega) = \begin{cases} \frac{1}{\sqrt{2}} |\hat{v}(js_k)|, & \omega \in [\omega_{k-1}, \omega_k] \cup [-\omega_k, -\omega_{k-1}], \\ 0 & \text{otherwise,} \end{cases}$$

where $s_k \in [\omega_{k-1}, \omega_k]$. Then it follows

$$\begin{aligned} \|\mathcal{S}_\tau v\|_{L_2} &\leq \left\| \mathcal{S}_\tau \circ \frac{1}{s} \right\|_{L_2} \cdot \|j\omega \cdot (\hat{v} - \hat{\underline{v}})\|_{L_2} + \|\mathcal{S}_\tau \underline{v}\|_{L_2} \\ &\leq \mathbf{O}(\|v - \underline{v}\|_{L_2}) + \sum_k \|\mathcal{S}_\tau \underline{v}_k\|_{L_2}, \end{aligned}$$

where $\mathbf{O}(\|v - \underline{v}\|_{L_2})$ can be made arbitrarily small. Now suppose we can find a bounded positive function $g(\omega)$ such that

$$\|\mathcal{S}_\tau \underline{v}_k\|_{L_2} \leq \left(\sup_{\omega \in [\omega_{k-1}, \omega_k]} g(\omega) \right) \|\underline{v}_k\|_{L_2}.$$

This would give

$$\begin{aligned} \|\mathcal{S}_\tau v\|_{L_2} &\leq \mathbf{O}(\|v - \underline{v}\|_{L_2}) + \left(\sup_{\omega \in [\omega_m, \omega_M]} g(\omega) \right) \sum_k \|\underline{v}_k\|_{L_2} \\ &\leq \mathbf{O}(\|v - \underline{v}\|_{L_2}) + \left(\sup_{\omega \in [\omega_m, \omega_M]} g(\omega) \right) \|v\|_{L_2}. \end{aligned}$$

To see the second inequality, we note that the Fourier transform of $\underline{v}_k$ is non-zero only on $[\omega_{k-1}, \omega_k] \cup [-\omega_k, -\omega_{k-1}]$. As such, $\sum_k \|\underline{v}_k\|_{L_2} = \|\sum_k \underline{v}_k\|_{L_2} = \|v\|_{L_2} + \mathbf{O}(\|v - \underline{v}\|_{L_2})$.

Function $\underline{v}_k(t)$ by definition is a sinc function of the following form:

$$\left(|\hat{v}(js_k)| \sqrt{\omega_k - \omega_{k-1}} \right) \cdot \frac{(\sin \omega_k t - \sin \omega_{k-1} t)}{\sqrt{\pi(\omega_k - \omega_{k-1})} t}$$

such that $\|\underline{v}_k(t)\|_{L_2} = |\hat{v}(js_k)|\sqrt{\omega_k - \omega_{k-1}}$. Therefore, the function $g(\omega)$ can be found by studying the upper bound of

$$\left\| \mathcal{S}_\tau \left(\frac{\sin \bar{\omega}t - \sin \underline{\omega}t}{\sqrt{\pi(\bar{\omega} - \underline{\omega})t}} \right) \right\|_{L_2},$$

where $0 \leq \underline{\omega} \leq \bar{\omega}$. We are especially interested in the case where $\bar{\omega}$ is arbitrarily close to $\underline{\omega}$. Let $\alpha := (\bar{\omega} + \underline{\omega})/2$ and $\beta := (\bar{\omega} - \underline{\omega})/2$. Let $f(t)$ be the sinc function

$$f(t) = \frac{\sin \bar{\omega}t - \sin \underline{\omega}t}{\sqrt{\pi(\bar{\omega} - \underline{\omega})t}}.$$

Note that the Fourier transform of $f(t)$ satisfies

$$\hat{f}(j\omega) = \begin{cases} \frac{1}{\sqrt{2(\bar{\omega} - \underline{\omega})}} & \text{if } \omega \in [\underline{\omega}, \bar{\omega}] \cup [-\bar{\omega}, -\underline{\omega}], \\ 0 & \text{otherwise.} \end{cases}$$

Furthermore, $f(t)$ can also be expressed as

$$f(t) = \frac{\sqrt{2} \cos(\alpha t) \sin(\beta t)}{\sqrt{\pi \beta t}} := \sqrt{2} \cos(\alpha t) f_b(t),$$

where the Fourier transform of $f_b(t)$ satisfies

$$\hat{f}_b(j\omega) = \begin{cases} \frac{1}{\sqrt{\bar{\omega} - \underline{\omega}}} & \text{if } \omega \in \left[-\frac{\bar{\omega} - \underline{\omega}}{2}, \frac{\bar{\omega} - \underline{\omega}}{2} \right], \\ 0 & \text{otherwise.} \end{cases}$$

It can be easily verified that $\|f\|_{L_2} = \|f_b\|_{L_2} = 1$.

We have the following expression for $\mathcal{S}_\tau f$:

$$\begin{aligned} & \frac{\sqrt{2} \cos(\alpha t) \sin(\beta t)}{\sqrt{\pi \beta t}} - \frac{\sqrt{2} \cos(\alpha(t - \tau)) \sin(\beta(t - \tau))}{\sqrt{\pi \beta(t - \tau)}} \\ & := f_1(t; \alpha, \beta) + f_2(t; \alpha, \beta) + f_3(t; \alpha, \beta), \end{aligned}$$

where $f_1(t; \alpha, \beta)$, $f_2(t; \alpha, \beta)$, and $f_3(t; \alpha, \beta)$ are, respectively, of the following forms

$$\sqrt{\frac{2}{\pi \beta}} (\cos(\alpha t) - \cos(\alpha(t - \tau))) \frac{\sin(\beta(t - \frac{\tau}{2}))}{t - \frac{\tau}{2}}, \quad (27)$$

$$\sqrt{\frac{2}{\pi \beta}} \cos(\alpha t) \left(\frac{\sin(\beta t)}{t} - \frac{\sin(\beta(t - \frac{\tau}{2}))}{t - \frac{\tau}{2}} \right), \quad (28)$$

$$\sqrt{\frac{2}{\pi \beta}} \cos(\alpha(t - \tau)) \left(\frac{\sin(\beta(t - \frac{\tau}{2}))}{t - \frac{\tau}{2}} - \frac{\sin(\beta(t - \tau))}{t - \tau} \right). \quad (29)$$

The following bounds on $f_i(t; \alpha, \beta)$ can be established.

Lemma 4. For any given α and β ,

$$|f_2(t; \alpha, \beta)| \leq \sqrt{\frac{8\beta}{\pi}}, \quad |f_3(t; \alpha, \beta)| \leq \sqrt{\frac{8\beta}{\pi}}, \quad \forall t. \quad (30)$$

Furthermore, there exist constants C_1 and C_2 such that

$$|f_2(t; \alpha, \beta)| \leq C_1 \sqrt{\beta} \frac{1}{|t|}, \quad |f_3(t; \alpha, \beta)| \leq C_2 \sqrt{\beta} \frac{1}{|t|}, \quad (31)$$

when t is sufficiently large.

Proof. Here we show the bounds for $f_2(t; \alpha, \beta)$. Those for $f_3(t; \alpha, \beta)$ are derived in exactly the same fashion. To see the inequality stated in (30), note that for any α, β , and t

$$\begin{aligned} |f_2(t; \alpha, \beta)| & \leq \left| \frac{\sqrt{2} \sin(\beta t)}{\sqrt{\pi \beta t}} - \frac{\sqrt{2} \sin(\beta(t - \frac{\tau}{2}))}{\sqrt{\pi \beta(t - \frac{\tau}{2})}} \right| \\ & \leq \sqrt{\frac{2\beta}{\pi}} \left| \frac{\sin(\beta t)}{\beta t} \right| + \sqrt{\frac{2\beta}{\pi}} \left| \frac{\sin(\beta(t - \frac{\tau}{2}))}{\beta(t - \frac{\tau}{2})} \right| \leq \sqrt{\frac{8\beta}{\pi}}. \end{aligned}$$

To show the inequality stated in (31), note that $f_2(t; \alpha, \beta)$ can be expressed as

$$\begin{aligned} & \sqrt{\frac{2\beta}{\pi}} \cos(\alpha t) \left(\frac{(1 - \cos(\beta \frac{\tau}{2})) \sin(\beta t)}{\beta(t - \frac{\tau}{2})} + \frac{\sin(\beta \frac{\tau}{2}) \cos(\beta t)}{\beta(t - \frac{\tau}{2})} \right. \\ & \quad \left. - \frac{\frac{\tau}{2} \sin(\beta t)}{\beta t(t - \frac{\tau}{2})} \right). \end{aligned}$$

Hence, for any given α and β , $|f_2(t; \alpha, \beta)| \leq \sqrt{2\beta/\pi}(T_1 + T_2 + T_3)$, where

$$T_1 = \left| \frac{(1 - \cos(\beta \frac{\tau}{2})) \sin(\beta t)}{\beta(t - \frac{\tau}{2})} \right| \leq \left| \frac{1 - \cos(\beta \frac{\tau}{2})}{\beta \frac{\tau}{2}} \right| \cdot \left| \frac{\frac{\tau}{2}}{t - \frac{\tau}{2}} \right|,$$

$$T_2 = \left| \frac{\sin(\beta \frac{\tau}{2}) \cos(\beta t)}{\beta(t - \frac{\tau}{2})} \right| \leq \left| \frac{\sin(\beta \frac{\tau}{2})}{\beta \frac{\tau}{2}} \right| \cdot \left| \frac{\frac{\tau}{2}}{t - \frac{\tau}{2}} \right|,$$

$$T_3 = \left| \frac{\frac{\tau}{2} \sin(\beta t)}{\beta t(t - \frac{\tau}{2})} \right| \leq \left| \frac{\sin(\beta t)}{\beta t} \right| \cdot \left| \frac{\frac{\tau}{2}}{t - \frac{\tau}{2}} \right|.$$

Therefore, all T_i 's are bounded by $|(\tau/2)/(t - \tau/2)|$, which in turn implies that

$$|f_2(t; \alpha, \beta)| \leq \frac{3\tau_0}{2} \sqrt{\frac{2\beta}{\pi}} \left| \frac{1}{t - \frac{\tau_0}{2}} \right|.$$

Here we use $0 \leq \tau(t) \leq \tau_0$ for all t . Thus, when t is sufficiently large, $|f_2(t; \alpha, \beta)| \leq C_1 \sqrt{\beta} \cdot |t|^{-1}$ for some constant C_1 . This concludes the proof. $\square$

For any α , let $h_{2,\beta}(t) := |f_2(t; \alpha, \beta)|^2$ and $h_{3,\beta}(t) := |f_3(t; \alpha, \beta)|^2$. Lemma 4 implies that

- Each $h_{i,\beta}(t)$, $i = 2, 3$, is upper bounded by an integrable function over $\mathbb{R}$.
- Each $h_{i,\beta}(t)$, $i = 2, 3$, converges to 0 pointwisely as β converges to 0.

By the Lebesgue dominated convergence theorem,

$$\lim_{\beta \rightarrow 0} \int_{\mathbb{R}} h_{i,\beta}(t) dt = \int_{\mathbb{R}} \lim_{\beta \rightarrow 0} h_{i,\beta}(t) dt = 0, \quad i = 2, 3.$$

In other words, for any α , $\|f_2(t; \alpha, \beta)\|_{L_2} \rightarrow 0$ and $\|f_3(t; \alpha, \beta)\|_{L_2} \rightarrow 0$ as $\beta \rightarrow 0$.

Now, let us consider $\|f_1(t; \alpha, \beta)\|_{L_2}$. Given α, β , one can express $f_1(t; \alpha, \beta)$ as $-2 \sin(\alpha \tau/2) f_p(t; \alpha, \beta)$, where

$$f_p(t; \alpha, \beta) = \sin \left(\alpha t - \alpha \frac{\tau}{2} \right) \left(\frac{\sqrt{2} \sin(\beta(t - \frac{\tau}{2}))}{\sqrt{\pi \beta(t - \frac{\tau}{2})}} \right).$$

Thus, $\|f_1(t; \alpha, \beta)\|_{L_2} \leq (\sup_t 2|\sin(\alpha\tau/2)|) \|f_p(t; \alpha, \beta)\|_{L_2}$. Furthermore, $\|f_p(t; \alpha, \beta)\|_{L_2}$ is bounded by

$$\begin{aligned} & \left(1 - \frac{d}{2}\right)^{-1} \int_{-\infty}^{\infty} \left(1 - \frac{\tau}{2}\right) \left(\sin \alpha \left(t - \frac{\tau}{2}\right)\right) \\ & \quad \times \frac{\sqrt{2} \sin(\beta(t - \frac{\tau}{2}))}{\sqrt{\pi\beta}(t - \frac{\tau}{2})} dt \\ & = \left(1 - \frac{d}{2}\right)^{-1} \int_{-\infty}^{\infty} \left(\sin(\alpha s) \frac{\sqrt{2} \sin(\beta s)}{\sqrt{\pi\beta}s}\right)^2 ds \\ & \leq \left(1 - \frac{d}{2}\right)^{-1}. \end{aligned}$$

Hence

$$\|f_1(t; \alpha, \beta)\|_{L_2} \leq \left(1 - \frac{d}{2}\right)^{-1/2} \sup_{\tau \in [0, h]} 2 \left|\sin\left(\alpha \frac{\tau}{2}\right)\right|.$$

Finally, $\|\mathcal{S}_\tau f\|_{L_2} \leq \|f_1(t; \alpha, \beta)\|_{L_2} + \|f_2(t; \alpha, \beta)\|_{L_2} + \|f_3(t; \alpha, \beta)\|_{L_2}$. Therefore, when $\beta \rightarrow 0$ (i.e., when $\underline{\omega} \rightarrow \bar{\omega}$), we have

$$\|\mathcal{S}_\tau f\|_{L_2} \leq \sup_{\tau \in [0, h]} 2 \left|\sin\left(\bar{\omega} \frac{\tau}{2}\right)\right| \sqrt{\frac{2}{2-d}} + \varepsilon,$$

where $\varepsilon = \mathbf{O}(|\bar{\omega} - \underline{\omega}|)$ can be made arbitrarily small. This concludes the proof.

References

- Cao, Y., Sun, Y., & Cheng, C. (1998). Delay-dependent robust stabilization of uncertain systems with multiple delay. *IEEE Transactions on Automatic Control*, 43(11), 1608–1612.
- Cervin, A., Henriksson, D., Lincoln, B., & Årzén, K. (2003). Analysis of control performance in real-time systems. *IEEE Transactions on Automatic Control*, 23(3), 16–30.
- Chen, J., & Latchman, H. A. (1995). Frequency sweeping tests for stability independent of delay. *IEEE Transactions on Automatic Control*, 40(9), 1640–1645.
- Cho, D., & Hedrick, J. K. (1989). Automotive powertrain modelling for control. *ASME Transaction*, 568–576.
- Fridman, E., & Shaked, U. (2002). An improved stabilization method for linear time-delay systems. *IEEE Transactions on Automatic Control*, 47(11), 1931–1937.
- Fridman, E., & Shaked, U. (2003). Delay-dependent stability and H_∞ control: Constant and time-varying delays. *International Journal of Control*, 76(1), 48–60.
- Fridman, E., & Shaked, U. (2005). Stability and L_2 -gain analysis of systems with time-varying delays: Input–output approach. In *Proceedings of the 44th IEEE conference on decision and control and the European control conference 2005*. Seville, Spain.
- Fu, M., Li, H., & Niculescu, S. I. (1998). Robust stability and stabilization of time-delay systems via integral quadratic constraint approach. In L. Dugard, E. I. Verriest (Eds.), *Stability and control of time-delay systems* (Vol. 228, pp. 101–115). Berlin: Springer.
- Gu, K., & Han, Q.-L. (2000). Discretized Lyapunov functional for linear uncertain systems with time-varying delay. In *Proceedings of the American control conference*.
- Gu, K., Kharitonov, V. L., & Chen, J. (2003). *Stability of time-delay systems*. Boston: Birkhäuser.
- Gu, K., & Niculescu, S.-I. (2000). Additional dynamics in transformed time-delay systems. *IEEE Transactions on Automatic Control*, 45(3), 572–575.
- Han, Q.-L., & Gu, K. (2001). On robust stability of time-delay systems with norm-bounded uncertainty. *IEEE Transactions on Automatic Control*, 46(9), 1426–1431.
- Huang, Y., & Zhou, K. (2000). Robust stability of uncertain time-delay systems. *IEEE Transactions on Automatic Control*, 45(11), 2169–2173.
- Jun, M., & Safonov, M. G. (2001). Iqc analysis for time-delay systems. *International Journal of Robust and Nonlinear Control*, 11, 1455–1468.
- Jun, M., & Safonov, M. G. (2002). Rational multiplier iqcs for uncertain time-delays and LMI stability conditions. *IEEE Transactions on Automatic Control*, 47(11), 1871–1875.
- Kao, C.-Y., & Lincoln, B. (2004). Simple stability criteria for systems with time-varying delays. *Automatica*, 40(8), 1429–1434.
- Kao, C.-Y., Megretski, A., Jönsson, U., & Rantzer, A. (2004). A MATLAB toolbox for robustness analysis. In *Proceedings of the 2004 IEEE conference on computer aided control systems design*. Taipei, Taiwan.
- Kao, C.-Y., & Rantzer, A. (2005). Robust stability analysis of linear systems with time-varying delays. In *Proceedings of the 16th IFAC world congress*. Prague, Czech Republic.
- Kharitonov, V. L., & Niculescu, S.-I. (2003). On the stability of linear systems with uncertain delay. *IEEE Transactions on Automatic Control*, 48(1), 127–132.
- Kim, J.-H. (2001). Delay and its time-derivative dependent robust stability of time-delayed linear systems with uncertainty. *IEEE Transactions on Automatic Control*, 46(5), 789–792.
- Kolmanovskii, V. B., Niculescu, S.-I., & Gu, K. (1999). Delay effects on stability: A survey. In *Proceedings of the 38th IEEE conference on decision and control*.
- Kolmanovskii, V. B., & Richard, J.-P. (1999). Stability of some linear systems with delays. *IEEE Transactions on Automatic Control*, 44(5), 984–989.
- Li, X., & de Souza, C. E. (1997a). Criteria for robust stability and stabilization of uncertain linear systems with state delay. *Automatica*, 33(9), 1657–1662.
- Li, X., & de Souza, C. E. (1997b). Delay-dependent robust stability and stabilization of uncertain linear delay systems: A linear matrix inequality approach. *IEEE Transactions on Automatic Control*, 42(8), 1144–1148.
- Low, S. H., Paganini, F., & Doyle, J. (2002). Internet congestion control. *IEEE Control Systems Magazine*, 22(1), 28–43.
- Megretski, A., & Rantzer, A. (1997). System analysis via integral quadratic constraints. *IEEE Transactions on Automatic Control*, 42(6), 819–830.
- Mehdi, D., Boukas, E. K., & Liu, Z.-K. (2002). Dynamical systems with multiple time-varying delays: Stability and stabilizability. *Journal of Optimization Theory and Applications*, 113(3), 537–565.
- Misra, V., Gong, W. B., & Towsley, D. (2002). Fluid-based analysis of a network of AQM routers supporting TCP-flows with an application to RED. In *Proceedings of ACM/SIGCOMM*.
- Morari, M., & Zafriou, E. (1989). *Robust process control*. Englewood Cliffs, NJ: Prentice-Hall.
- Niculescu, S.-I. (2001). Delay effects on stability. A robust control approach. *Lecture notes in control and information sciences*. Heidelberg: Springer.
- Park, P. (1999). A delay-dependent stability criterion for systems with uncertain time-invariant delays. *IEEE Transactions on Automatic Control*, 44(4), 876–877.
- Suplin, V., Fridman, E., & Shaked, U. (2006). H_∞ control of linear uncertain time-delay systems—a projection approach. *IEEE Transactions on Automatic Control*, 51(4), 680–685.
- Wang, Z. Q., Lundström, P., & Skogestad, S. (1994). Representation of uncertain time delays in the H -infinity framework. *International Journal of Control*, 59(3), 627–638.
- Wu, M., He, Y., She, J.-H., & Liu, G.-P. (2004). Delay-dependent criteria for robust stability of time-varying delay systems. *Automatica*, 40(8), 1435–1439.
- Xu, S., & Lam, J. (2005). Improved delay-dependent stability criteria for time-delay systems. *IEEE Transactions on Automatic Control*, 50(3), 384–387.
- Zhang, J., Knope, C. R., & Tsotras, P. (2001). Stability of time-delay systems: Equivalence between Lyapunov and scaled small-gain conditions. *IEEE Transactions on Automatic Control*, 46(3), 482–486.

Zhang, J., Knope, C. R., & Tsiotras, P. (2005). New results on delay-dependent control of time-delay systems. *IEEE Transactions on Automatic Control*, 50(1), 95–100.

Zhang, W., Branicky, M. S., & Phillips, S. M. (2001). Stability of networked control systems. *IEEE Control Systems Magazine*, 21(1), 84–99.



Chung-Yao Kao was born in Taiwan. He received the Sc.D. degree in mechanical engineering from Massachusetts Institute of Technology, Cambridge, MA, USA, in 2002. From 2002 to 2004, he held research positions at the Department of Automatic Control of the Lund Institute of Technology, Lund, Sweden, the Mittag-Leffler Institute, Stockholm, Sweden, and the Division of Optimization and Systems Theory at the Royal Institute of Technology, Stockholm, Sweden. He is currently a senior lecturer in the Department of Electrical and Electronic Engineering at the University of Melbourne.

His research interests include analysis and design of networked and distributed systems, specialized computational algorithms for control system analysis and synthesis, and control system applications.



Anders Rantzer was born in 1963. He received a Ph.D. degree from KTH, Stockholm, Sweden. After postdoctoral positions at KTH and at University of Minnesota, he joined Lund University in 1993. In 1999, he was appointed as a professor of Automatic Control.

Prof. Rantzer was a winner of the 1990 SIAM Student Paper Competition and the 1996 IFAC Congress Young Author Price. He is a Fellow of IEEE and has served as associate editor of *IEEE Transactions on Automatic Control* and several other journals.

His research interests are in modelling, analysis, and synthesis of control systems, with particular attention to robustness, optimization, and distributed control.